impop_k hands-on tutorial

IBS, IBD, and local ancestry from pangenome alignments

MEMPANG workshop

Time budget. 45 minutes. Compute time across the entire tutorial is a few seconds; the rest of the time is reading outputs, plotting, and answering interpretation questions. Work alone or in pairs.

Setup assumptions. You have the `workshop/` directory on disk (cloned from the course repository), the four pre-built binaries in `workshop/bin/`, the precomputed identity matrices in `workshop/data/`, and Python 3 with `numpy` and `matplotlib`. All commands below *must* be run from the `workshop/` directory.

Where things came from. Yesterday you ran `impg similarity` on PAF alignments to obtain pairwise identity per window. The two TSVs in `data/` are exactly that — precomputed on `chr12`, so you don't pay the cost again today. The reproduction script is `code/regenerate_data.sh`.

Why chr12 long arm. Every section below works on the long (q) arm of `chr12` — roughly 40 to 130 Mb — staying clear of the centromere at ∼34–38 Mb and of both telomeres. The arm gives us enough length to resolve recombination breakpoints visually.

1. Setup (3 min)

```
$ cd workshop
$ ls bin/ data/ code/
$ ./bin/ancestry -help | head -3
```

You should see four binaries (`ibs`, `ibd`, `ancestry`, `jacquard`), TSV/text files in `data/`, and Python/Bash scripts in `code/`. The third command should print three lines of help text starting with "Local ancestry inference".

If something fails

If `./bin/ancestry -help` reports a missing library, your VM's `glibc` is older than the one the binaries were built against (2.39). Tell the instructor; the fix is to copy the binaries from a Linux laptop with a more recent `glibc` or to build them from source (`cargo build -release` in the impop repository).

2. Part 1 — IBS: looking at identity windows (7 min)

The dataset

`data/identity_chr12_pedigree.tsv` contains pairwise identity per 10-kb window for the CEPH 1463 platinum pedigree on `chr12:40-130 Mb` (the long q arm). Eighteen haplotypes total: five full siblings (NA12879, NA12881, NA12882, NA12885, NA12886), the two paternal grandparents (NA12889, NA12890), and the two maternal grandparents (NA12891, NA12892). The two grandparental couples are mutually unrelated.

```
$ head -3 data/identity_chr12_pedigree.tsv
$ wc -l data/identity_chr12_pedigree.tsv
```

Columns: `chrom`, `start`, `end`, `group.a`, `group.b`, `estimated.identity`, `group.a.length`, `group.b.length`. The `group` columns are haplotype identifiers in `<sample>#<hap>` format. About 1.4M pair-window rows.

Run the explorer

```
$ python3 code/01_explore_ibs.py
```

This loads the TSV, splits the rows into intra-individual pairs (e.g. `NA12879#1` vs. `NA12879#2`) and inter-individual pairs (everything else), and writes `figures/student_ibs_histogram.png`. Open it.

1.1

The right tail of the histogram concentrates at identity ≥ 0.9995. What biological scenarios produce values that high? Name two distinct ones.

1.2

In this pedigree-only dataset, the intra-individual and inter-individual means are extremely close, and the inter-individual mean is in fact *slightly higher*. That is counter-intuitive. Why does it make sense for this cohort? *Background you need:* the intra-individual class is hap1-vs-hap2 of the *same* person — two largely unrelated homologs. The inter-individual class includes every sibling–sibling and sibling–grandparent pair in the cohort, and those pairs share large IBD tracts (∼50% and ∼25% respectively). Use that to explain the direction of the gap.

1.3

The histogram is plotted on `[0.985, 1.001]`, but the raw `estimated.identity` column contains values as low as ∼0.12. From what you know about pangenome alignment to a linear reference, what produces values that low? (Hint: these are not biology; they are technical.)

3. Part 2 — IBD detection (12 min)

If "HMM" is new

The IBD detector is a 2-state hidden Markov model: each window is in state `IBD` or `not-IBD`, transitions between adjacent windows are calibrated by recombination rate, and the emission $P(\text{identity} \mid \text{state})$ is learned by Baum–Welch from the data. The flags below (`-p-enter-ibd`, `-baum-welch-iters`) directly control these two HMM ingredients. See the talk slide "HMM in 30 seconds" for the picture.

What we expect to find

Full siblings share, on average, 50% of the genome IBD. On a single 90 Mb arm we therefore expect several multi-Mb IBD segments per sibling pair. Between **each sibling and their paternal grandparents** (NA12889/NA12890) the expected fraction is 25%; same between each sibling and their maternal grandparents (NA12891/NA12892). Between the two grandparental couples, expected IBD is essentially zero.

Run the IBD detector

```
$ bash code/02_run_ibd.sh
```

This invokes `bin/ibd` on the pedigree identity matrix and writes `solutions/ibd_segments.tsv`. The relevant flags:

```
-similarity-file data/identity_chr12_pedigree.tsv
-region chr12:40000001-130000000
-size 10000
-min-len-bp 2000000
-min-lod 3.0
-baum-welch-iters 20
```

- `-min-len-bp 2000000`: minimum reported segment length (2 Mb). Standard value for an arm-scale scan.
- `-min-lod`: minimum LOD score $\log_{10}[P(\text{data} \mid \text{IBD}) \,/\, P(\text{data} \mid \neg\text{IBD})]$. A value of 3 means the segment is $10^3\times$ more likely under IBD.
- `-baum-welch-iters 20`: EM iterations for emission parameter learning.

Read the output

```
$ head -20 solutions/ibd_segments.tsv
```

Columns: `chrom, start, end, group.a, group.b, n_windows, mean_identity, mean_posterior, ..., lod_score`. Each row is one detected IBD segment between one haplotype pair.

2.1

Find one IBD segment between a pair of siblings (any of NA12879, NA12881, NA12882, NA12885, NA12886 with another). Write down the interval and the LOD score. Bonus: find a pair with more than one multi-Mb IBD segment on this arm — how many breakpoints are implied?

2.2

There are detected "IBD" segments between `NA12892#1` and `NA12892#2` — the two haplotypes of one person — and also between the two haplotypes of *other* grandparents and even of some siblings. Strict IBD between the two haplotypes of one diploid individual would imply autozygosity. What conservative explanation should you consider first, and what would you check to discriminate?

2.3

The HMM has two key transition parameters that you did not tune: `-p-enter-ibd` (default 10^{-4}) and `-expected-seg-windows` (default 50). What would you expect to happen to the count of detected segments if you set `-p-enter-ibd` to 10^{-2}? What about `-expected-seg-windows` to 5?

4. Part 3 — Local ancestry inference (16 min)

How the chimeras and the panel were built

The five chimeric query haplotypes were stitched from real HPRCv2 assemblies. Each `CHIM_NN#1` is a mosaic of one AFR donor and one EUR donor, copied along ground-truth tracts:

chimera	AFR donor	EUR donor	ground-truth tracts
CHIM_00#1	HG01884	HG00097	`data/ground_truth_tracts.tsv`
CHIM_01#1	HG01884	HG00099	...
CHIM_02#1	HG01884	HG00126	...
CHIM_03#1	HG01884	HG00128	...
CHIM_04#1	HG01884	HG00133	...

The reference panel in `data/populations.tsv` is a different set of 20 HPRCv2 haplotypes: 10 AFR (HG02257, HG02258, HG02280, HG02451, HG02572, both `#1` and `#2`) and 10 EUR (HG00140, HG00146, HG00232, HG00253, HG00272, both haps each). **None of the panel samples are donors of the chimeras.** It is a fair test: the model has to recognize the donor's population without ever having seen the donor.

The dataset

`data/identity_chr12_admix.tsv` contains pairwise identity per 10-kb window for `chr12:50-120 Mb` — 70 Mb in the middle of the long q arm. Because 70 Mb is much longer than a typical admixture tract, each chimera carries 3–7 ground-truth breakpoints in this window.

Run local ancestry inference

```
$ bash code/03_run_ancestry.sh
```

The relevant flags:

```
-similarity-file data/identity_chr12_admix.tsv
-query-samples data/queries.txt
-populations data/populations.tsv
-region chr12:50000001-120000000
-window-size 10000
-auto-configure
-identity-floor 0.9
```

- `-auto-configure`: estimates two emission-model knobs from the data, instead of asking you to set them.
  - *Softmax temperature* T: rescales per-window similarity differences before turning them into emission probabilities via $\text{softmax}(\Delta/T)$. Small $T \to$ sharp commit to one population from tiny similarity gaps; large $T \to$ smooth, conservative posteriors.
  - *Pairwise-contrast weight*: how much weight the emission gives to the *difference* between max-similarity-to-population-A and max-similarity-to-B, relative to the absolute identity. High contrast weight emphasizes which panel is closer; low contrast weight makes the emission depend more on raw identity values.
- `-identity-floor 0.9`: drops windows whose maximum similarity across the panel is below 0.9 — usually alignment-gap artifacts.

Inspect the painting

```
$ head -20 solutions/ancestry_segments.tsv
$ python3 code/04_plot_painting.py
```

This produces `figures/student_ancestry_painting.png` with two bars per chimera: "truth" on top, "impop" below. Open it.

3.1

Pick one chimera and count its ground-truth breakpoints in the 50–120 Mb interval. How many of them does the model recover within ∼1 Mb? Which ones does it miss or hallucinate?

3.2

Scan the five paintings side by side. Are the errors *the same kind* across chimeras, or chimera-specific? A recurring error in roughly the same genomic region across multiple, independent chimeras suggests a property of the *region*, not of the chimeras. Do you see one?

3.3

Look at the `mean_posterior` column of the TSV. Several wrong calls carry posteriors ≥ 0.9. How can the model be confidently wrong? *Hint: think about the difference between the model being confident in its assumptions and the model being correct against reality.*

Open the black box: forward-backward, Viterbi, and the raw input

The TSV you just read is the *Viterbi* decoding: the single most likely sequence of ancestry states. The HMM also computes a *marginal posterior* per window via forward-backward: the probability of each state at each window, integrating over all paths. The script invoked `ancestry` with `-posteriors-output`, so you already have these per-window numbers in `solutions/ancestry_posteriors.tsv`.

```
$ head -3 solutions/ancestry_posteriors.tsv
$ python3 code/07_plot_posteriors.py
```

This produces two figures:

- `figures/student_posterior_trace.png` for `CHIM_03#1`: ground truth on top, P(AFR) per window in the middle (the marginal posterior), Viterbi-decoded segments at the bottom. The dashed line at $P = 0.5$ marks where Viterbi flips.
- `figures/student_panel_similarity.png`: the raw input the HMM emissions see. Per window, Δ = (max identity to AFR panel) − (max identity to EUR panel), smoothed over 250 kb. Notice how small the contrast is — median $|\Delta| \approx 1\text{–}3 \times 10^{-4}$ per single window. The HMM has to chain dozens of windows to make a confident call from such a weak per-window signal.

3.4

Open `student_posterior_trace.png`. Find a window where P(AFR) sits between 0.4 and 0.6: the model is genuinely uncertain there. What is Viterbi's call at that same window, and what does that tell you about the difference between the marginal posterior (forward-backward) and the most-likely-path (Viterbi)?

5. Part 4 — Grandparent painting of the CEPH pedigree (7 min)

We swap roles. Instead of a population panel, we use the *four grandparents* as four "populations" (K=4), and paint each grandchild's two haplotypes against them. This is the same HMM, same binary, same identity matrix as Part 2 — only the panel file changes.

The setup

```
$ cat data/pedigree_populations.tsv
$ cat data/pedigree_queries.txt
```

Four "populations", each containing the two haplotypes of one grandparent:

```
GP_PAT NA12889#1, NA12889#2 (paternal grandfather)
GM_PAT NA12890#1, NA12890#2 (paternal grandmother)
GP_MAT NA12891#1, NA12891#2 (maternal grandfather)
GM_MAT NA12892#1, NA12892#2 (maternal grandmother)
```

The ten queries are the ten haplotypes of the five grandchildren (NA12879, NA12881, NA12882, NA12885, NA12886 — `#1` and `#2` each).

Run the painting

```
$ bash code/05_paint_pedigree.sh
$ python3 code/06_plot_pedigree_painting.py
```

This produces `figures/student_pedigree_painting.png`: two stacked bars per grandchild, one per haplotype, coloured by inferred grandparent of origin.

What you should see

The opening slide of the talk ("a chromosome inherited through a pedigree") is now a concrete result on real data. In particular:

- Each grandchild's two haplotypes split cleanly: one is a mosaic of *only* GP_PAT + GM_PAT blocks (paternal homolog), the other of *only* GP_MAT + GM_MAT blocks (maternal homolog).
- Breakpoints within a homolog mark actual meiotic recombinations in the mother or father of the CEPH family.
- The VCF never told us which haplotype index (`#1` or `#2`) was paternal. The painting does.

4.1

For one grandchild of your choice, tell me which of `#1` and `#2` is paternal and which is maternal. Justify from the figure, not from the FASTA headers.

4.2

Pick a grandchild with ≥ 2 breakpoints on one homolog in the 40–130 Mb window. How many generations of meiotic recombination do those breakpoints represent (think about which parent generated them)?

4.3

If the model ever placed a GP_MAT block on the same homolog as a GP_PAT block, that would be biologically impossible — a grandchild cannot receive one homolog from both father and mother. Do you see any such violations in your figure? If yes, what would be the first thing you would check before blaming the biology?

Hand-in (optional)

If your instructor asks: write a 1-paragraph summary covering (a) one thing you found in the IBS histogram that was not obvious before, (b) one IBD segment you can confidently call between two named haplotypes, (c) one ancestry call from Part 3 where the model is wrong with high confidence and what that tells you about calibration, and (d) which grandchild haplotype indices (`#1` vs `#2`) you concluded to be paternal vs maternal from Part 4.